

 L&T Energy Hydrocarbon	Indicative Inspection & Test Plan for Desuperheater		
Quality Control and Inspection	AdVENT-STD-QC-ITP-050	Rev.01	25-June-25

Indicative Inspection & Test Plan for DESUPERHEATER



					
25-June-25	01	Issued for Implementation	SD	VVK	RVS
Date	Rev.	Description	Prepared By	Reviewed By	Approved By

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Revision Record

Rev. No.	Date	Chapter / Clause	Revision Note
1	25-June-25	All	Revised Issue
0	06-Sept-23	All	First Issue

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Contents

Revision Record	2
1 Scope	4
2 Reference Documents	4
3 Definitions	4
4 Notes	4
5 Indicative Inspection & Test Plan	5

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1 Scope

This inspection and test plan covers the minimum testing requirements of the **Desuperheater**. The scope of inspection shall be as per PO / Contractual requirements. The Sub-contractor / manufacturer / vendor / supplier shall prepare stage wise inspection & testing plan in details complying the requirements of applicable specification, material requisition & data sheets and shall submit for review/approval.

2 Reference Documents

- Material Requisition or Purchase Requisition
- Purchase Order
- Code & Standards
- Project Specifications & Design Philosophy.

3 Definitions

LTEH	L&T Energy Hydrocarbon (Purchaser)
TPIA	Third Party Inspection Agency
Perform Point "P"	Manufacturer / Supplier shall 100% perform inspection check.
Hold Point "H"	Hold for Inspection and do not proceed. Fabrication shall be kept under hold until inspection has been performed or written clearance is obtained by Vendor. The work cannot proceed unless the Hold Point is cleared.
Witness Point "W"	Witness for Inspection Advise of inspection, Fabrication may be continued if inspector is not present after invitation issued in due time.
Review Point "R"	Review / Check / Verify documents. Review of Procedures / Test Reports / Certificates / Inspection Reports
Approval Point "A"	Approval shall be obtained on documents.

4 Notes

- This is an INDICATIVE INSPECTION & TEST PLAN (IITP) only. Vendor to cover all major stages in line with data sheet, drawings, relevant MR, statutory requirements (if any) and governing code guidelines.
- Detailed INSPECTION & TEST PLAN shall be submitted for LTEH approval.
- In the event of any conflict between the technical requirements of a project specification, procedure, or drawing, the most stringent requirement shall govern.
- Acceptance norms for all the activities shall be as per PO / purchase requisition / code & standards referred therein / project specifications / approved documents.
- Vendor shall carry out additional testing required, if any not specified in above, as per PO / specification.

5 Indicative Inspection & Test Plan										
Item No	Activity	Extent of supply	Reference Document	Acceptance Criteria	Verifying Document	Inspection Agency				Remarks
						Vendor	TPIA	LTEH	OWNER/PMC	
1	Review & Approval of Detailed Inspection & Test Plan.	100%	Approved MR, Datasheet, GAD's and applicable Specs/Standards.	Approved MR, Datasheet, GAD's and applicable Specs/Standards.	ITP	P	R	A	A	
1.1	NDT / QA-QC Procedures: DPT, MT, RT, UT Hydrotest, Pneumatic test, Heat Treatment procedures etc. (As applicable)	100%	API 598 ASME B 16.34 ASME Section V	API 598 ASME B 16.34 ASME Section V	Approved NDT Procedures	H	R	R	R	Duly approved by NDT level III person.
1.2	WPS / PQR / WPQ for welded valves, Overlay, Stellite & Weld repair (As applicable)	100%	ASME Section IX	ASME Section IX	Approved WPS, PQR & WPQ	H	R	R	R	Duly certified by Reputed TPIA
2	Material Inspection									
2.1	a) Material Identification and review of MTC for - 3.1 Mill test certificates as per EN 10204 for Valve Body, Bonnet, Plug, Seat, closure member & trim material (Spray Head, Nozzle & Spindle), Bar/ Forging & Fasteners. 2.2 test certificate as per EN 10204 for non-pressure containing parts. b) IGC Test for all SS Material (As per approved datasheet)	100%	Approved MR, Datasheet, GAD's and applicable Specs ASTM/ASME/Standards.	Approved MR, Datasheet, GAD's and applicable Specs ASTM/ASME/Standards.	MTC's / Lab report / Heat Treatment Charts	H	R	R	R	

Item No	Activity	Extent of supply	Reference Document	Acceptance Criteria	Verifying Document	Inspection Agency				Remarks
						Vendor	TPIA	LTEH	OWNER/ PMC	
3	In Process Inspection-									
3.1	PT/MT & Thickness verification for overlay & hard facing.	100%	Approved MR, Datasheet, GAD's and applicable Specs/Standards.	Approved MR, Datasheet, GAD's and applicable Specs/Standards.	Inspection Report / NDT Reports, RT Films	H	W*/R	R	R	Review of RT films, RT, UT, PT reports to be provided for review for 100%. *10% witness by TPIA.
3.2	RT of Butt weld Joint (As applicable) - soundness	100%					W*/R	R	R	
3.3	LPT/PT of Fillet weld Joint (As applicable)-soundness	100%					W*/R	R	R	
3.4	UT for Forging, Bar stock (Trim Parts- Plug, Seat & Spindle etc.] (above Ø50 mm)	100%					W*/R	R	R	
3.5	Physical & Visual Checks which includes: Markings, Overall dimensions, Face to face body dimension, Flanges verification, Thickness check, Checks for accessories, connections & installations. Functional checks on complete valve with all accessories.	100%			Inspection Report	H	W	R	R	
4	PMI Test									
4.1	PMI Test - All types of valves, rupture disks, bodies, pipe, flanges & trim (Applicable for SS & AS material), bonnets & covers (as applicable)	100%	Approved MR, Datasheet, GAD's and applicable Specs/ Standards.	Approved MR, Datasheet, GAD's and applicable Specs/ Standards.	PMI reports	P	W*	R	R	100% PMI - review *10% PMI Witness
5	Final Inspection									
5.1	Hydrostatic Shell Test: -At 1.5 times of max. Operating pressure.	100%	Approved MR, Datasheet, GAD's and applicable Specs/ Standards. API-598	Approved MR, Datasheet, GAD's and applicable Specs/ Standards. API-598	Hydro test reports	P	RW	RW	R	

Item No	Activity	Extent of supply	Reference Document	Acceptance Criteria	Verifying Document	Inspection Agency				Remarks
						Vendor	TPIA	LTEH	OWNER/PMC	
5.2	Hydrostatic Seat leakage test	100%	Approved MR, Datasheet, GAD's and applicable Specs/ Standards. API-598	Approved MR, Datasheet, GAD's and applicable Specs/ Standards. API-598	Hydro test reports	P	RW	RW	R	
5.3	Pneumatic Leakage Test	100%			Pneumatic test reports	P	RW	RW	R	
5.4	Visual inspection: Verification of type and model, Tag no, Nameplate details, Paint shade, Accessories, Quantity, Marking (nominal size, material of flange and direction of flow), Workmanship.	100%	Approved MR, Datasheet, GAD's and applicable Specs/Standards.	Approved MR, Datasheet, GAD's and applicable Specs/Standards.	Inspection reports	P	W	W	R	
5.5	Harmful defects: 1. Defect such as cracks, deformation and flaws shall not be found in the casting, forging and machined surface of the pressure rating part. 2. Defect such as inside surface weld protrusion; lack of fusion and incomplete penetration shall not be found in welded places of pressure retaining part.	100%								
5.6	Dimension Inspection: Complete dimension as per the approved GAD.	100%								
5.7	Performance Check: Functional test of complete assembly (as applicable). Functional/Calibration Test on Valve, Positioner/Robotor, Actuator & Hand wheel & accessories mounted. (Stroke, Linearity & Hysteresis) Paint shade & DFT check	100%								

Item No	Activity	Extent of supply	Reference Document	Acceptance Criteria	Verifying Document	Inspection Agency				Remarks
						Vendor	TPIA	LTEH	OWNER/PMC	
6	Spares / Loose Items (as per PO) Visual, Quantity, Make, Model, Rating, Dimension (as applicable)	100%	As per PO annexure/ Approved BOM or BOQ	As per PO annexure/ Approved BOM or BOQ	Spare offer list	P	W	R	R	Refer Note-6
7	Documents Review									
7.1	1. Manufacturer's test certificates for all bought out items, accessories & spares etc. 2. Material co-relation chart showing Make, Model number, Sr Number, MTC/TC Number, Heat number, Statutory Certificate (As Applicable). Calibration & Performance Certificates, NDT, Hydrotest, IBR certificates. other TCs & Certificates as required by Data Sheets and Proj. Specs.	100%	Approved MR, Datasheet, GAD's and applicable Specs/Standards.	Approved MR, Datasheet, GAD's and applicable Specs/Standards.	MTC, TC's / reports	P	R	R	R	
8	Inspection Release Note (IRN) / Certificate including Spares	100%	Compliance with PO / MR requirements	Compliance with PO / MR requirements	IRN	H		H	H	

Notes: -

1. In the event of conflict between purchase specifications, contract documents and ITP, conditions that are more stringent shall be applicable.
2. This document generally describes the requirements pertaining to all types of the item. Requirements specific to PO and the item are only applicable.
3. Acceptance norms for all the activities shall be as per PO/PR/Standards referred there in/Job specification/Approved documents.
4. Vendors shall carry out additional testing required, if any not specified above, as per PO/Technical specification.
5. Calibration status of measuring and monitoring instruments/machines to be verified by TPIA during inspection activities.
6. Spares shall be offered for inspection along with main Items - "Separate Annexure with Main IRN" or "Separate IRN" for Spares shall be obtained in line with approved Spare list.
7. Wherever "W/R" Is mentioned, it is at the sole discretion of "LTEH" to Decide "W" or "R" while approval of ITP during order execution.
8. Inspection stages of Owner/PMC/Licenser would be informed during order execution/Pre-Inspection meeting.
9. For Sub-Vendor Items, the main vendor shall be involved in the inspection prior to offering to LTEH/TPIA.
10. All statutory approval / certification shall be taken by vendor wherever applicable.
11. For AS & SS Material IGC test as per ASTM Practice "E" to be carried out, Additionally, Microscopic structure to be observed at 250X magnification for checking no carbide precipitation.